

Hands-on Lab: Create a Firewall Rule in Microsoft Windows Defender



**Skills
Network**

Estimated time needed: **30 minutes**

About This Lab

This exercise will look at Windows Defender Firewall with Advanced Security. This advanced view provides more in-depth options for configuration. All Windows Firewall rules, and their details, are stored here, allowing you to edit configurations for each rule or exception.

Objectives

In this hands-on lab, you will:

- Use Windows Defender Firewall with Advanced Security to edit an existing firewall rule.
- Enforce the following rules:
- Allow the connection for Key Management Service on the Domain and Private network.
- Deny the connection for Key Management Service on the Public network.

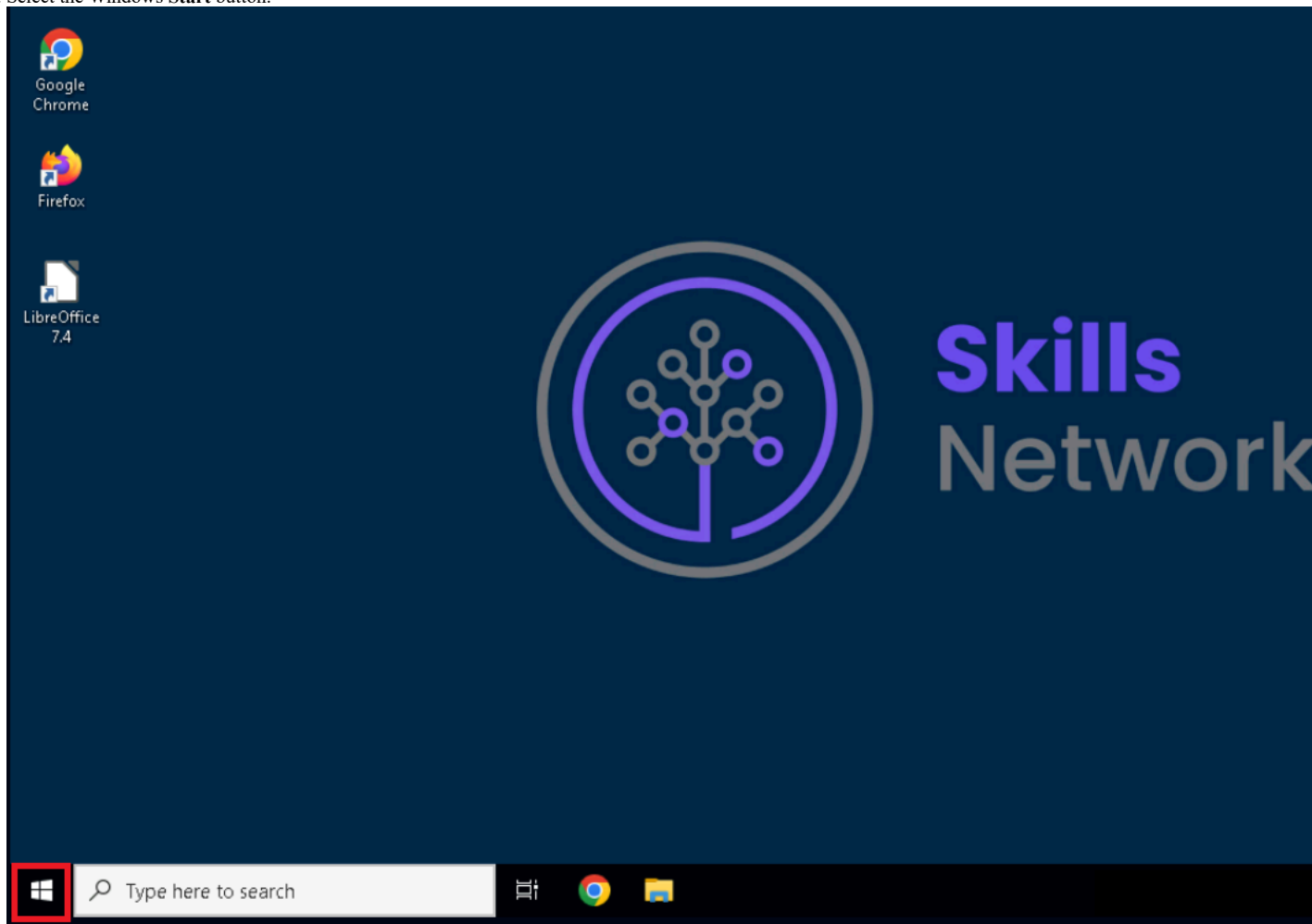
Important Information About Lab Instructions and Solutions

In case you try to use your physical keyboard in the lab environment, it might not produce any visible results. To avoid this issue, please use the On-Screen Keyboard (you can find it by searching for On-Screen Keyboard in the search bar at the bottom of your screen). If search functionality doesn't work, you can also click on the Windows icon, scroll down to find Windows Ease of Access, click on it, and then select On-Screen Keyboard.

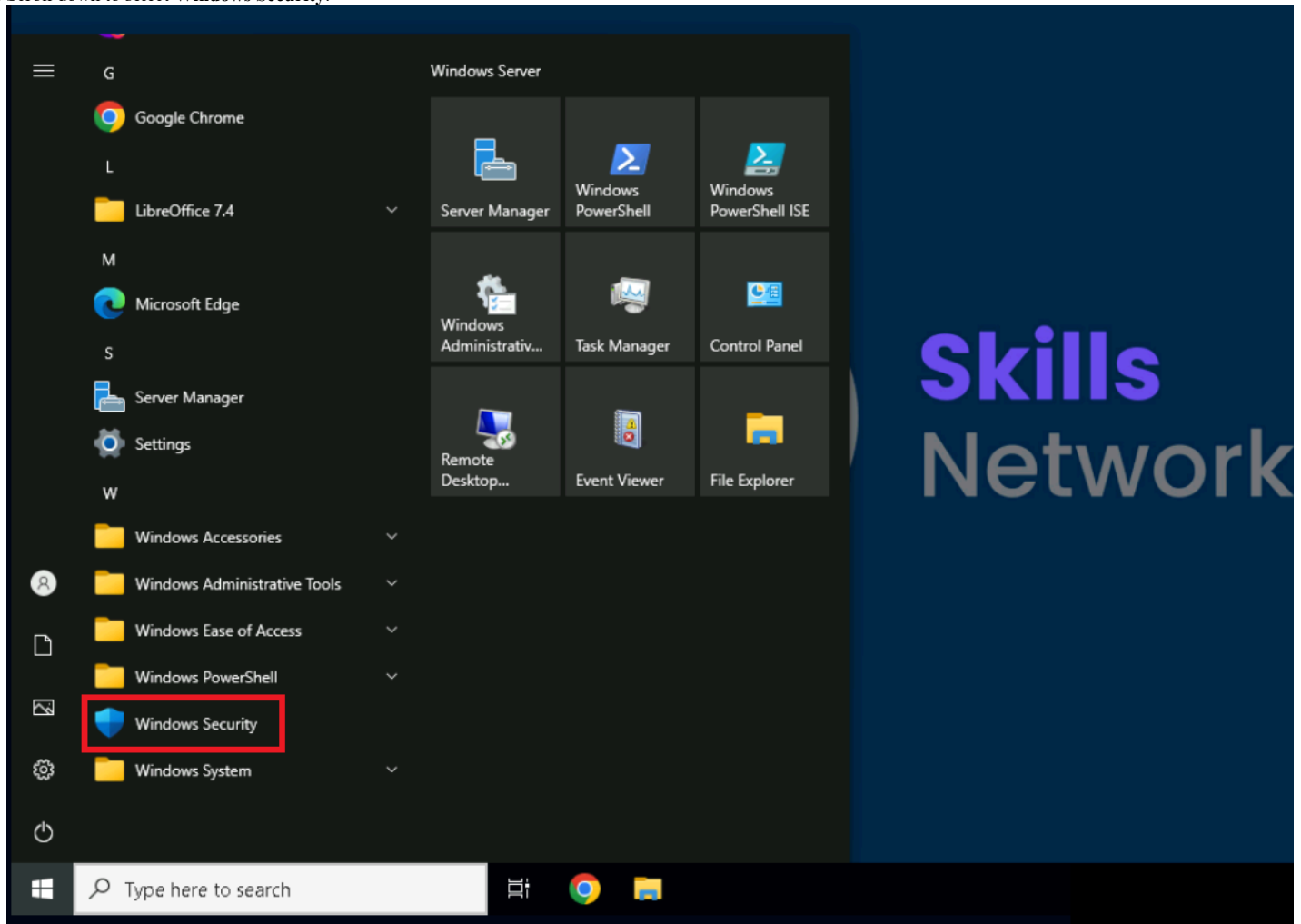
Microsoft Windows operating system features can vary based on the Windows edition. If completing these exercises on your machine, your navigation and solutions may differ from what's presented in this lab.

Exercise 1: Allow an app through a firewall

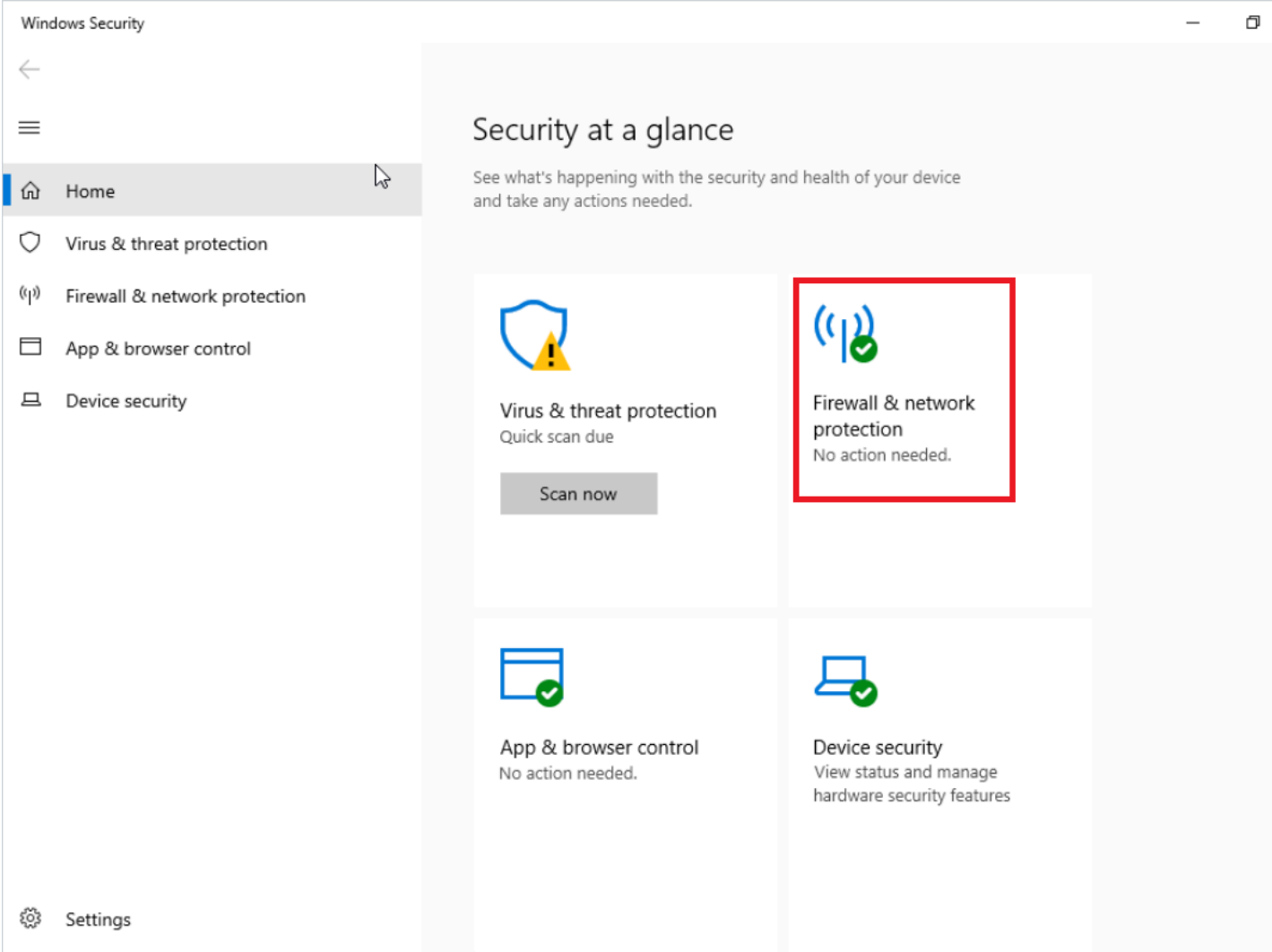
1. Select the Windows **Start** button.



2. Scroll down to select **Windows Security**.



3. Select **Firewall & network protection**.



4. Select **Allow an app through firewall**.

Windows Security

←

☰

🏠 Home

🛡️ Virus & threat protection

🔒 Firewall & network protection

📁 App & browser control

📱 Device security

Firewall & network protection

Who and what can access your networks.

🏢 **Domain network**

Firewall is on.

🏠 **Private network**

Firewall is on.

🌐 **Public network (active)**

Firewall is on.

Allow an app through firewall

[Network and Internet troubleshooter](#)

[Firewall notification settings](#)

[Advanced settings](#)

[Restore firewalls to default](#)

🔧 Settings

Windows Community videos

[Learn more about Firewall & network protection](#)

Who's protecting me?

[Manage providers](#)

Change your privacy settings

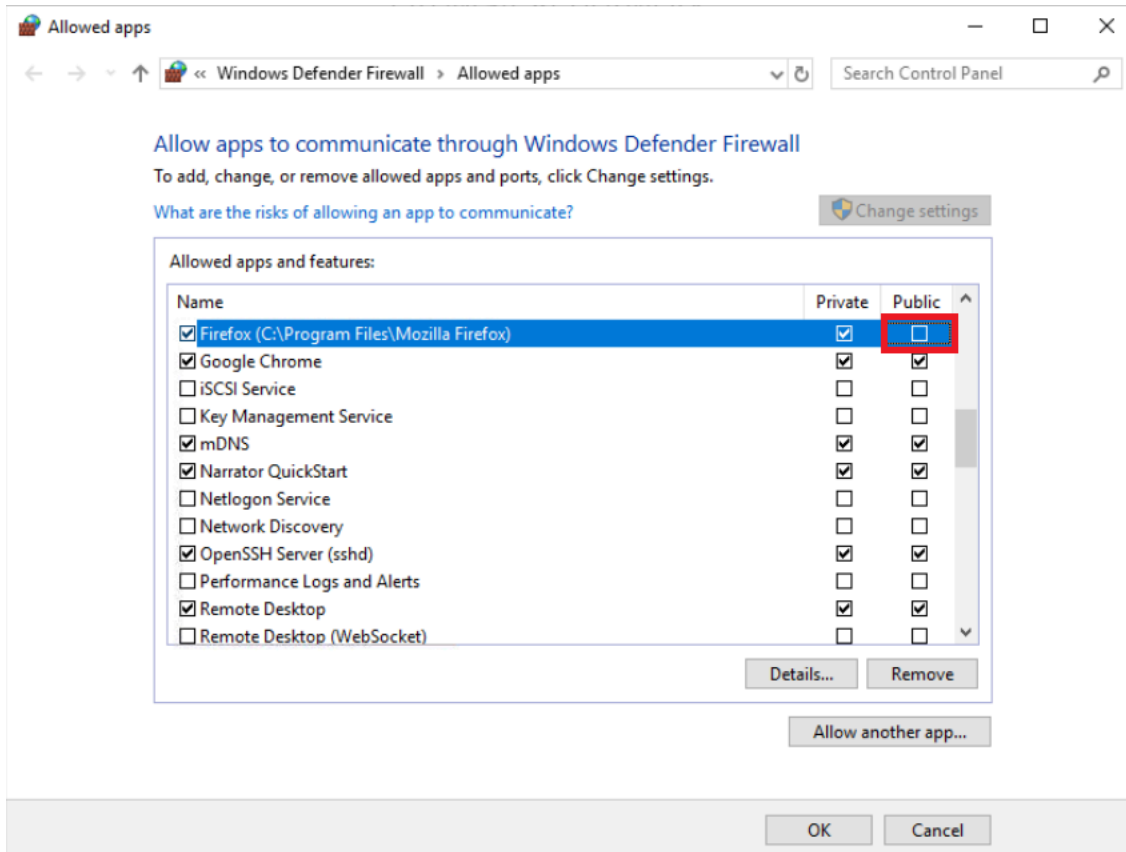
View and change privacy settings for your Windows 10 device.

[Privacy settings](#)

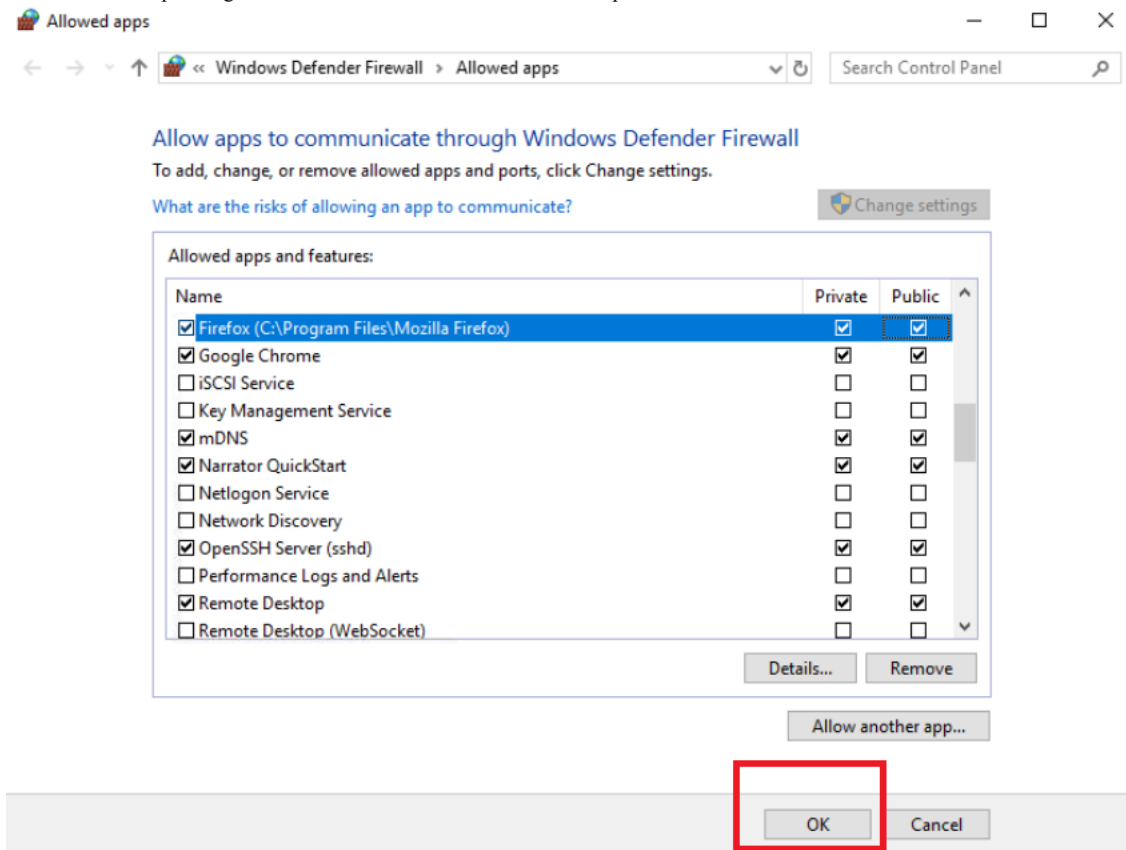
[Privacy dashboard](#)

[Privacy Statement](#)

5. Here you will see a list of allowed apps and features. Scroll down to Firefox (C:\Program Files\Mozilla Firefox). Notice that communication is permitted on the private, but not the public. Select the box to enable communication on the public network.

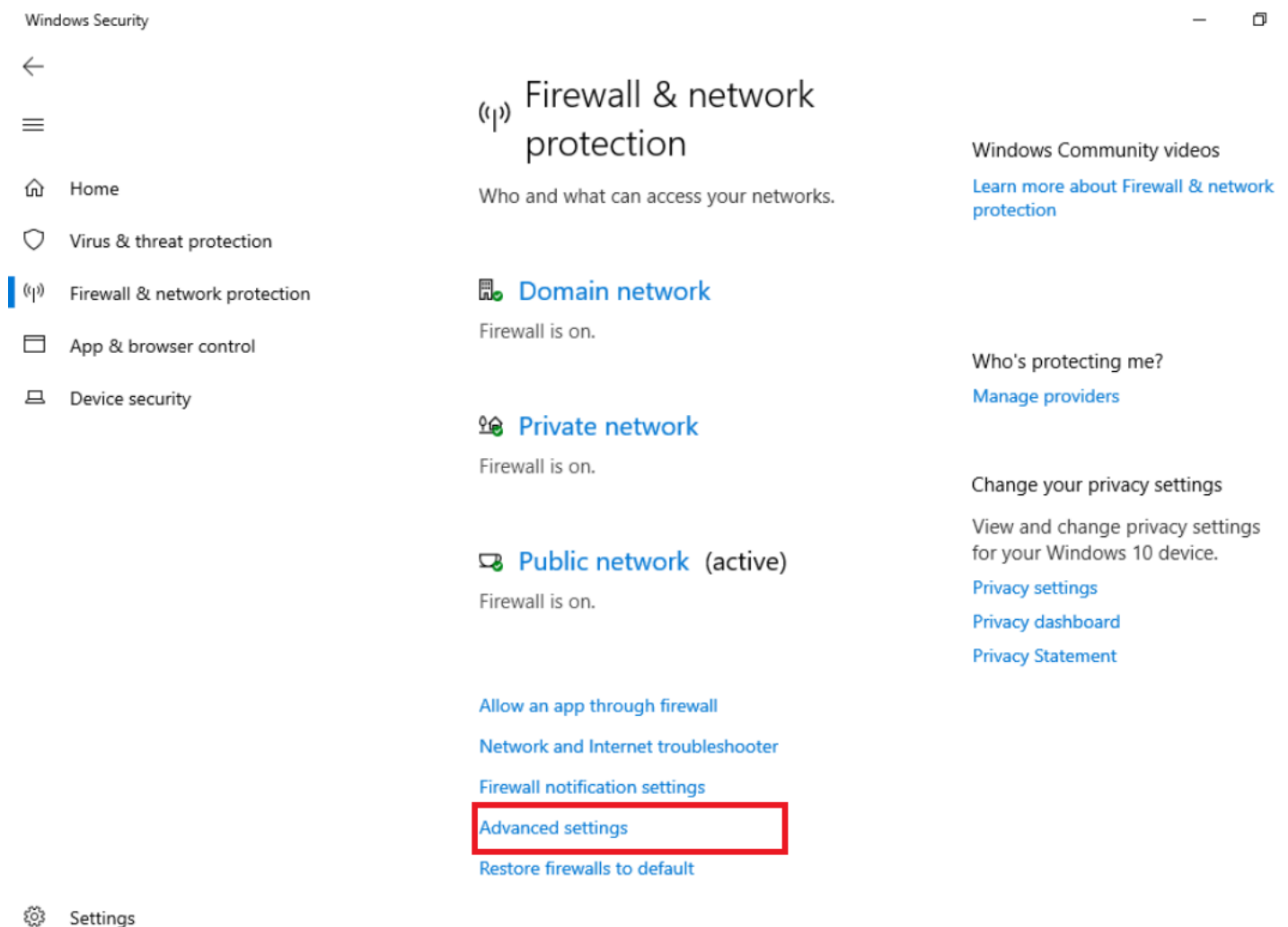


6. Select **OK** to accept changes and to return to the Firewall and network protection screen.



Exercise 2: Enable Inbound Rules to Allow Remote Service Management

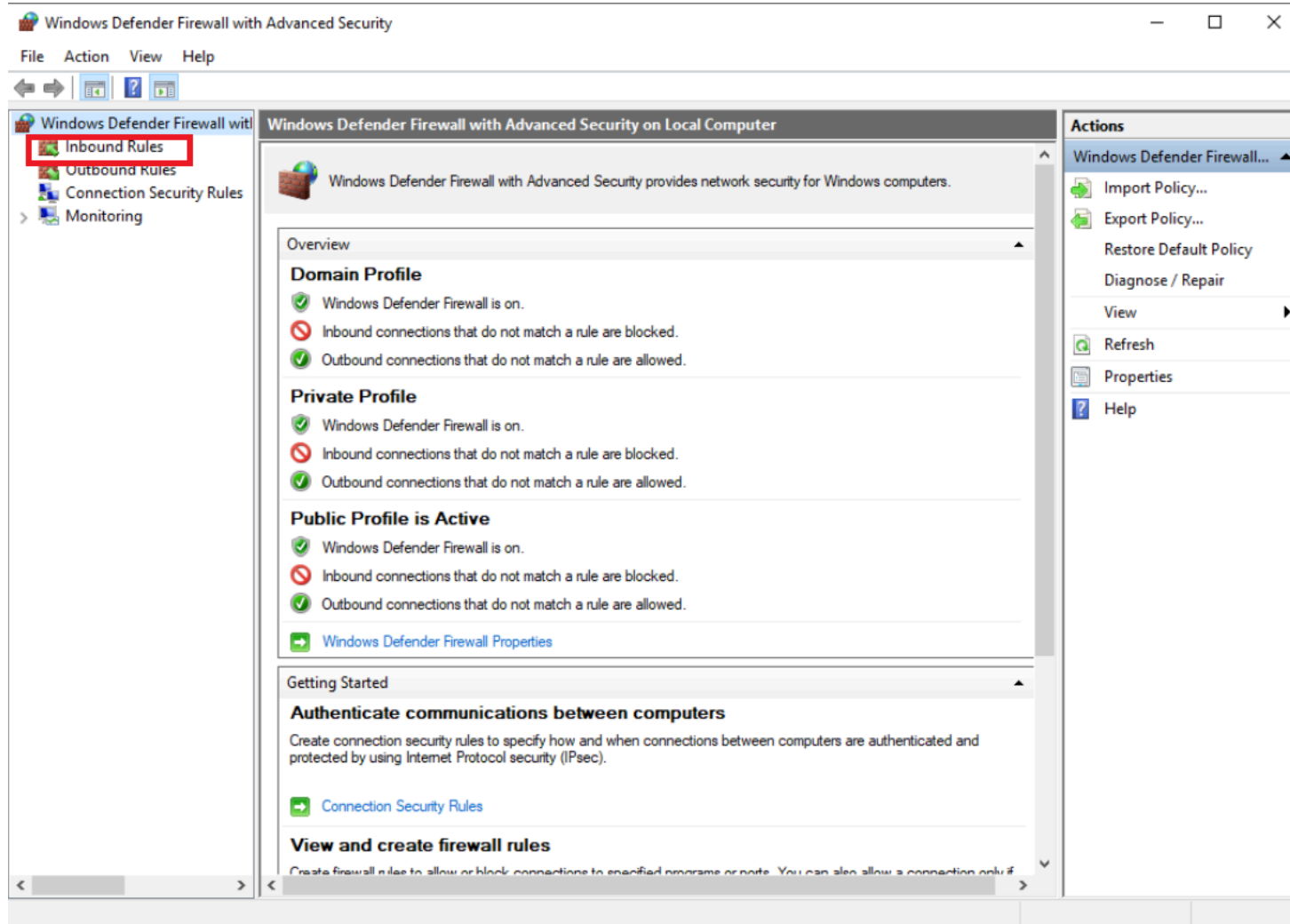
1. Select **Advanced settings** on the Firewall & network protection screen.



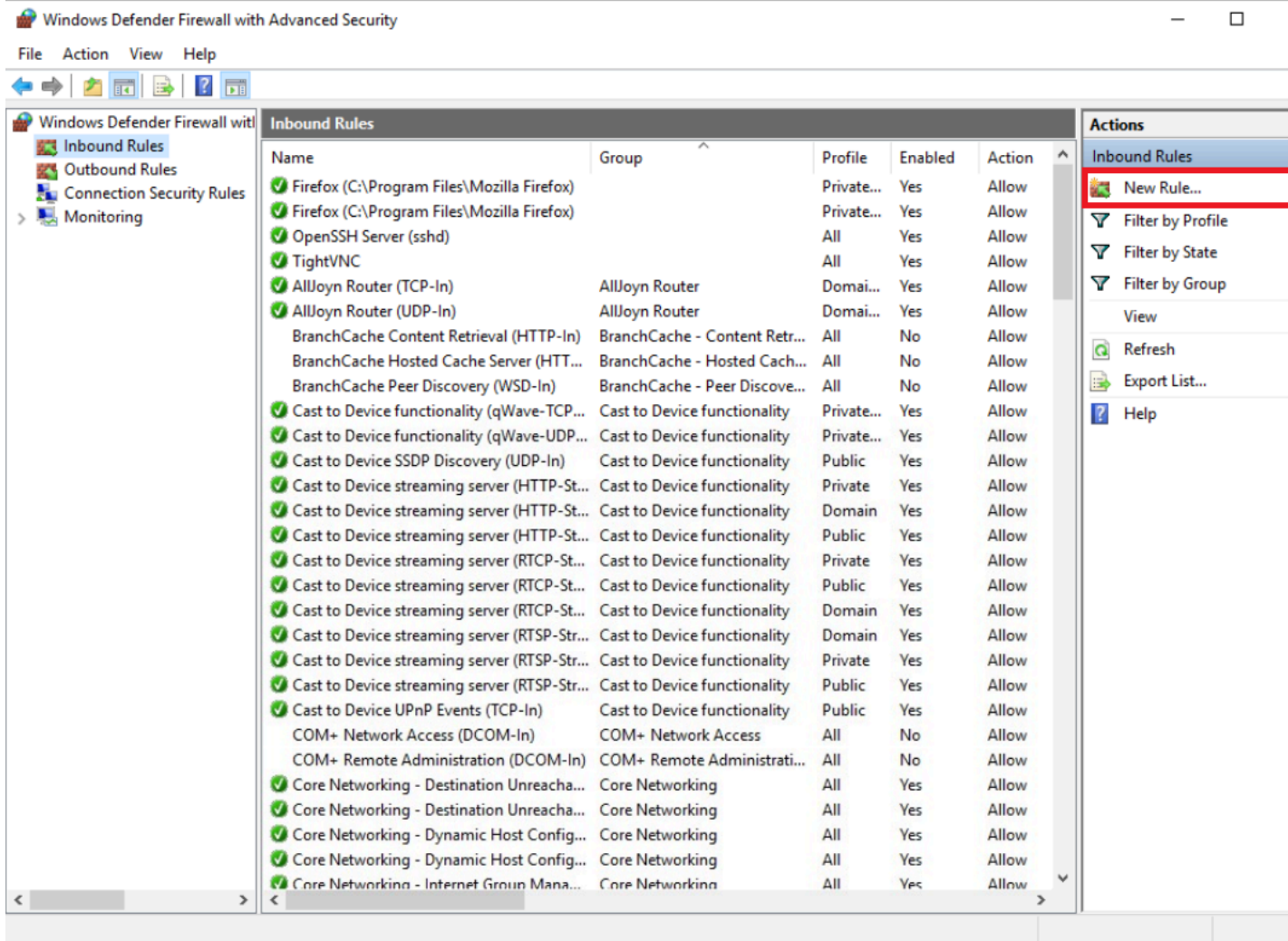
2. Here you will see an **Overview** in the center panel. On the left side, you will see three different rule types:

- Inbound rules
- Outbound rules
- Connection security rules

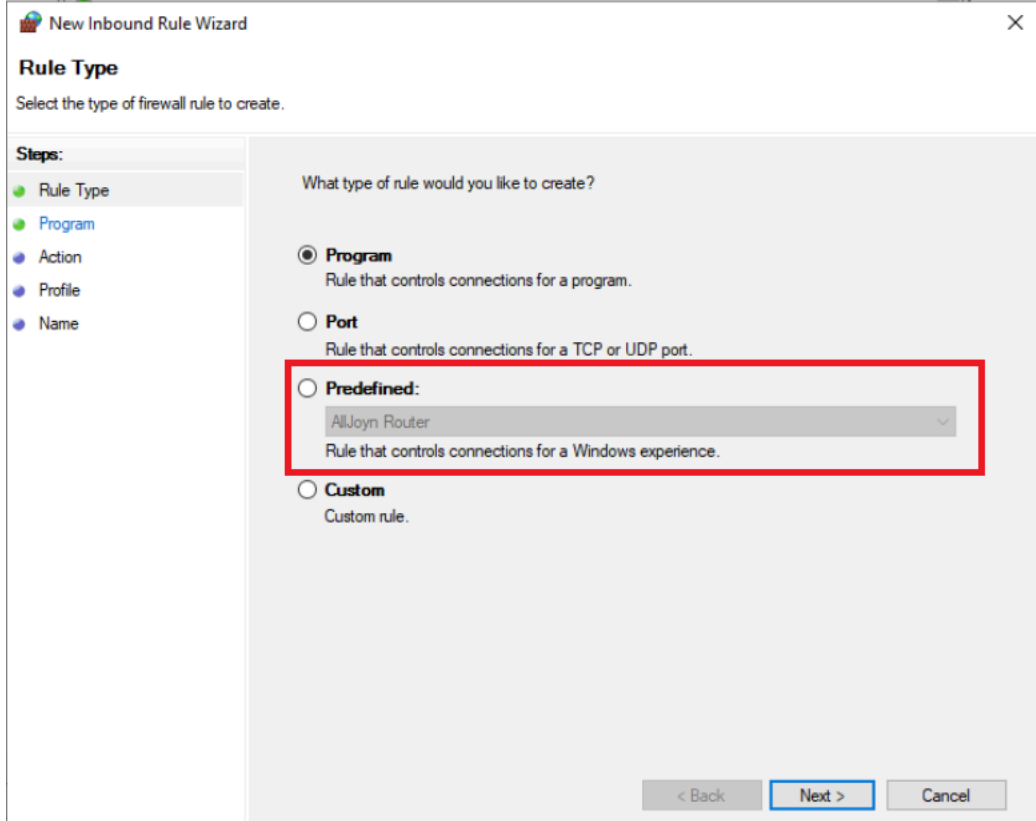
These rules can be configured to filter traffic based on computers, users, applications, ports, protocols, etc. Select **Inbound rules**.



3. Select **New Rule** in the right pane.



4. Here you will see options for four new rule types. Select **Predefined**.



5. Select **Remote Service Management** from the dropdown list.

New Inbound Rule Wizard

Rule Type

Select the type of firewall rule to create.

Steps:

- Rule Type
- Predefined Rules
- Action

What type of rule would you like to create?

Performance Logs and Alerts

Remote Desktop

Remote Desktop (WebSocket)

Remote Event Log Management

Remote Event Monitor

Remote Scheduled Tasks Management

Remote Service Management

Remote Shutdown

Remote Volume Management

Routing and Remote Access

Secure Socket Tunneling Protocol

SNMP Trap

Software Load Balancer

TPM Virtual Smart Card Management

Windows Defender Firewall Remote Management

Windows Management Instrumentation (WMI)

Windows Media Player

Windows Media Player Network Sharing Service

Windows Media Player Network Sharing Service (Internet)

Windows Remote Management

Windows Remote Management (Compatibility)

Remote Shutdown

Rule that controls connections for a Windows experience.

Custom

Custom rule.

< Back

Next >

Cancel

6. Select **Next** to continue.

New Inbound Rule Wizard

Rule Type

Select the type of firewall rule to create.

Steps:

- Rule Type
- Predefined Rules
- Action

What type of rule would you like to create?

Program

Port

Predefined:

Remote Service Management

Custom

Rule that controls connections for a program.

Rule that controls connections for a TCP or UDP port.

Rule that controls connections for a Windows experience.

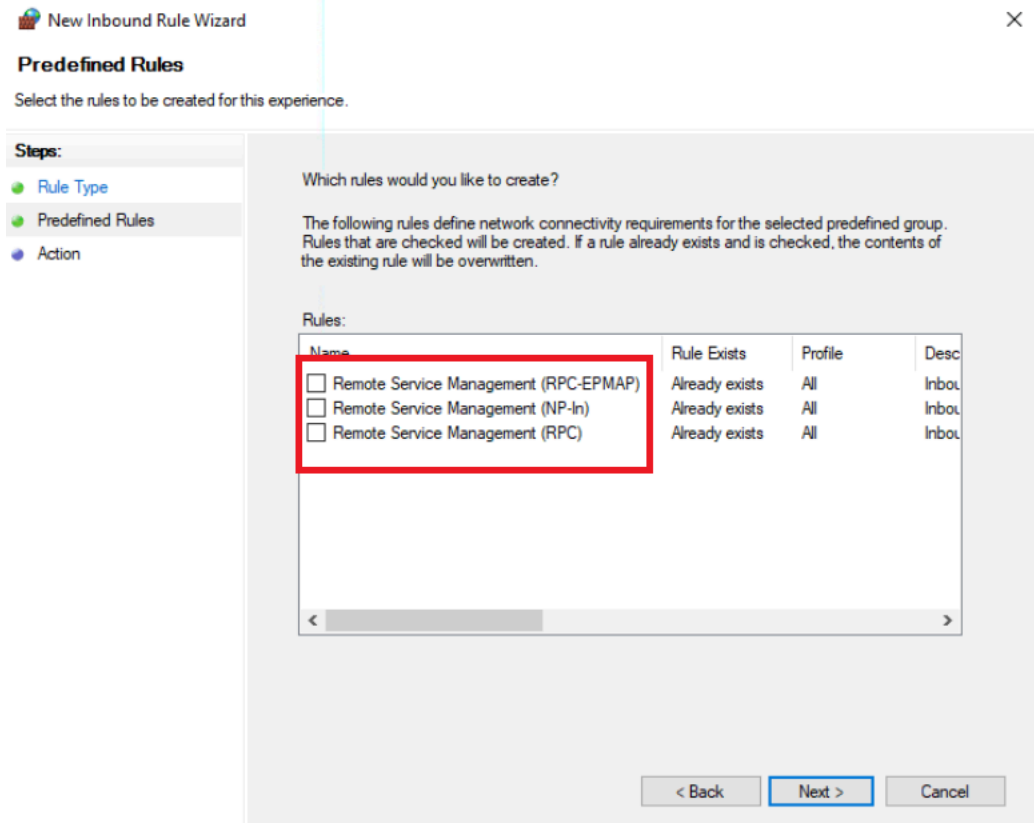
Custom rule.

< Back

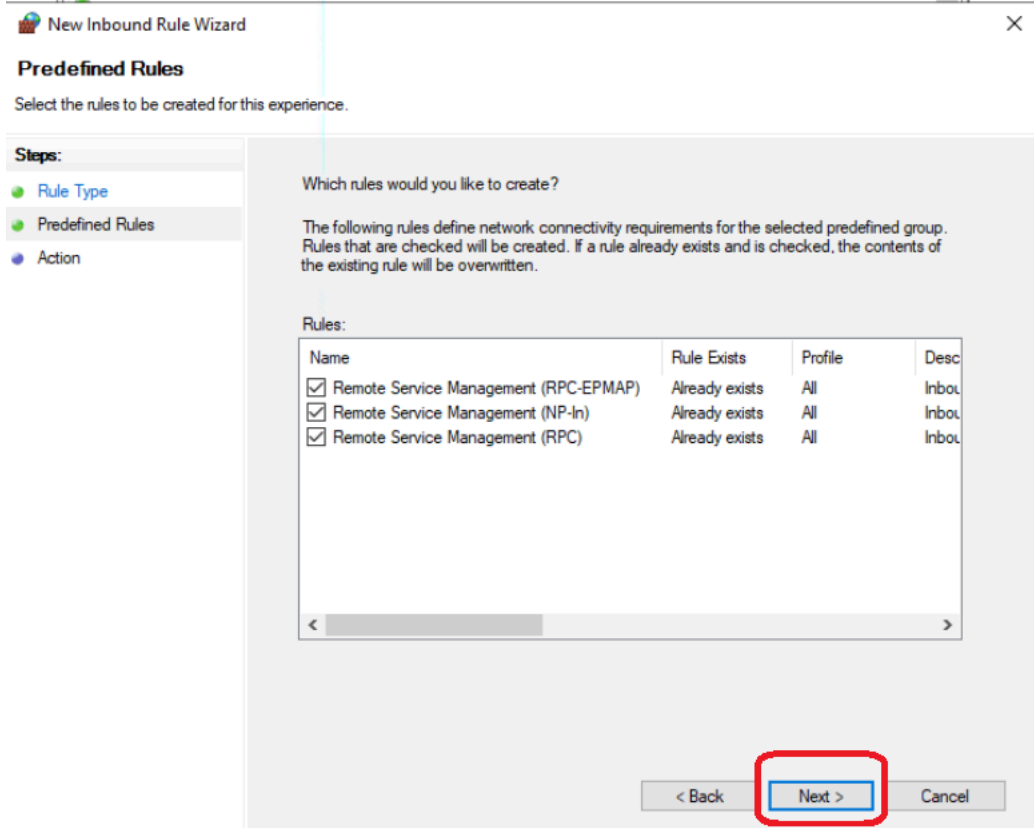
Next >

Cancel

7. Check the boxes for **Remote Service Management (RPC-EPMAP)**, **Remote Service Management (NP-IN)**, and **Remote Service Management (RPC)**.



8. Select **Next** to continue.



9. Select **Allow the connection if it is secure**.

New Inbound Rule Wizard

Action

Specify the action to be taken when a connection matches the conditions specified in the rule.

Steps:

Rule Type

Predefined Rules

Action

What action should be taken when a connection matches the specified conditions?

☒ **Allow the connection**
This includes connections that are protected with IPsec as well as those are not.

☐ **Allow the connection if it is secure**
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.

Customize...

☐ **Block the connection**

< Back

Finish

Cancel

10. Select **Next**.

New Inbound Rule Wizard

Action

Specify the action to be taken when a connection matches the conditions specified in the rule.

Steps:

Rule Type

Predefined Rules

Action

Users

Computers

What action should be taken when a connection matches the specified conditions?

☐ **Allow the connection**
This includes connections that are protected with IPsec as well as those are not.

☒ **Allow the connection if it is secure**
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.

Customize...

☐ **Block the connection**

< Back

Next >

Cancel

11. Verify that the box to **only allow connections from these users** has been checked. Select **Add** to add an authorized user.

The screenshot shows the 'New Inbound Rule Wizard' window, specifically the 'Users' step. The title bar reads 'New Inbound Rule Wizard' with a close button. The main heading is 'Users' with the instruction 'Specify the users that are allowed to make the connection specified by this rule.' On the left, a 'Steps:' pane lists 'Rule Type', 'Predefined Rules', 'Action', 'Users' (highlighted), and 'Computers'. The main area has two sections: 'Authorized users' and 'Exceptions'. In the 'Authorized users' section, the checkbox 'Only allow connections from these users' is checked and highlighted with a red box. To its right, an 'Add...' button is also highlighted with a red box. Below this is an empty list box and a 'Remove' button. The 'Exceptions' section has an unchecked checkbox 'Skip this rule for connections from these users', an empty list box, and 'Add...' and 'Remove' buttons. A note at the bottom states: 'Note: user identities can only be verified if an authentication method that carries user identity is used.' At the bottom right are '< Back', 'Next >' (highlighted with a blue border), and 'Cancel' buttons.

12. Enter "Administrator" into the **Enter the object names to select** box.

The screenshot shows the 'Select Users or Groups' dialog box. The title bar reads 'Select Users or Groups' with a close button. It contains several fields and buttons: 'Select this object type:' with a dropdown showing 'Users, Groups, or Built-in security principals' and an 'Object Types...' button; 'From this location:' with a dropdown showing 'EC2AMAZ-J5GP1U8' and a 'Locations...' button; 'Enter the object names to select (examples):' with an empty text box highlighted by a red rectangle; and a 'Check Names' button. At the bottom are 'Advanced...', 'OK', and 'Cancel' buttons.

13. Select **OK**.

This screenshot shows the 'Select Users or Groups' dialog box after the text 'Administrator' has been entered into the 'Enter the object names to select (examples):' text box. The 'OK' button at the bottom is highlighted with a red rectangle. All other elements, including the title bar, dropdowns, and other buttons, are the same as in the previous screenshot.

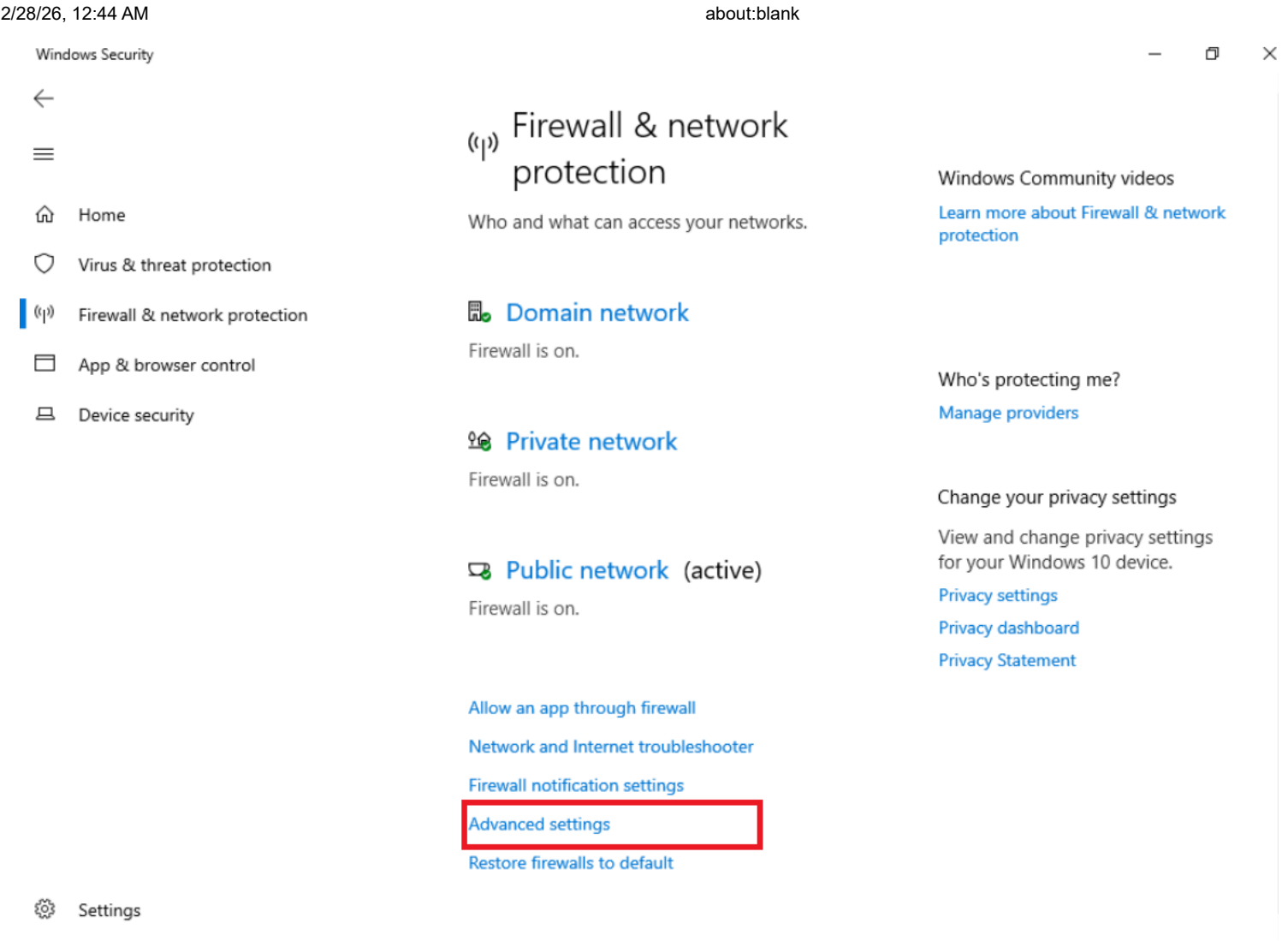
14. Click **Next** and Select **Finish**.

The screenshot shows the 'New Inbound Rule Wizard' window, specifically the 'Computers' step. The window has a title bar with a close button (X) and a subtitle 'New Inbound Rule Wizard'. Below the title bar, the word 'Computers' is displayed in bold. A subtitle reads: 'Specify the computers that are allowed to make the connection specified by this rule.' On the left side, there is a 'Steps:' panel with a list of steps: 'Rule Type', 'Predefined Rules', 'Action', 'Users', and 'Computers'. The 'Computers' step is currently selected and highlighted. The main area of the wizard is divided into two sections: 'Authorized computers' and 'Exceptions'. The 'Authorized computers' section has a checkbox labeled 'Only allow connections from these computers:' which is currently unchecked. Below this checkbox is a large empty rectangular box for listing authorized computers, with 'Add...' and 'Remove' buttons to its right. The 'Exceptions' section has a checkbox labeled 'Skip this rule for connections from these computers:' which is also unchecked. Below this checkbox is another large empty rectangular box for listing exceptions, with 'Add...' and 'Remove' buttons to its right. At the bottom of the wizard, there is a note: 'Note: computer identities can only be verified if an authentication method that carries computer identity is used.' At the very bottom, there are three buttons: '< Back', 'Finish', and 'Cancel'. The 'Finish' button is highlighted with a red rectangular border.

Exercise 3: Allow Key Management Service on the Domain and Private network, and deny the connection on the Public network

A KMS is used to activate Microsoft products (such as Windows and Office) within an organization without requiring each machine to connect directly to Microsoft for activation.

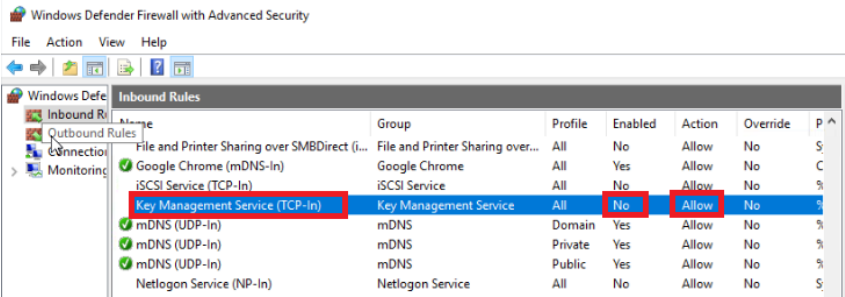
1. Open the Windows Defender Firewall with Advanced Security options



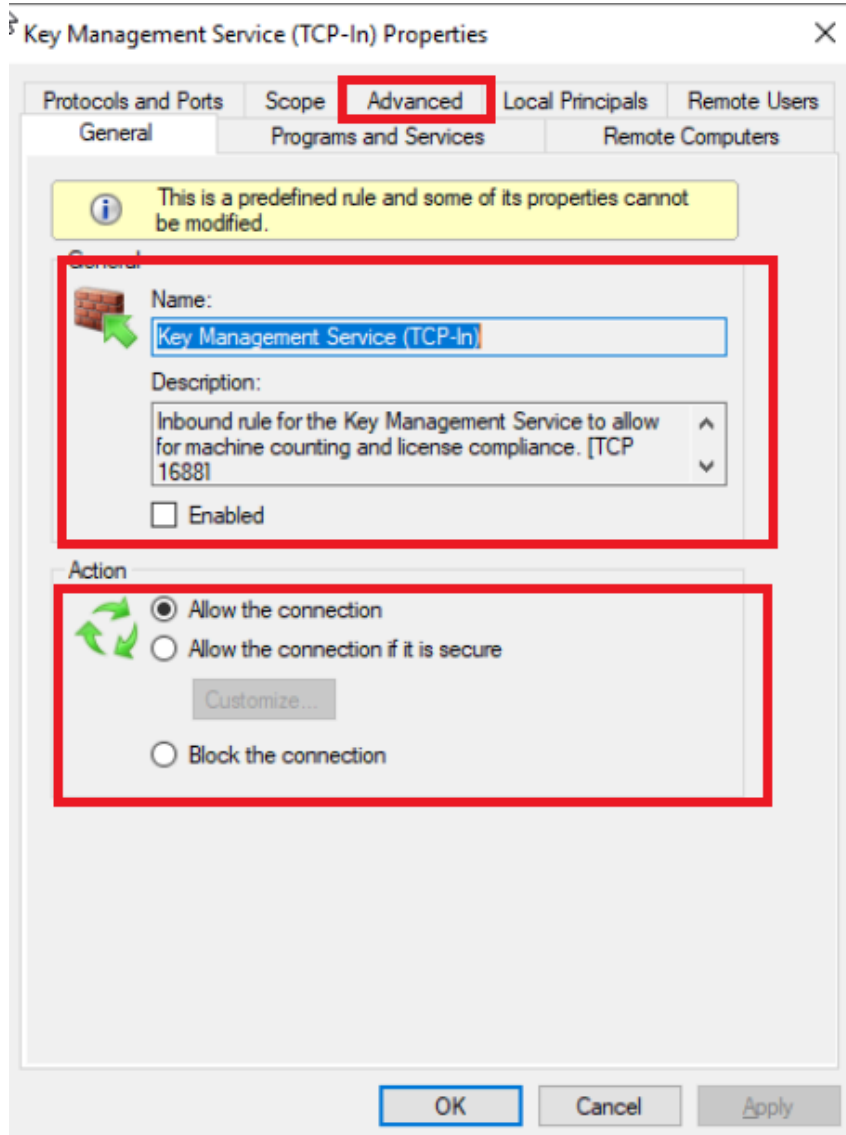
2. Scroll to the **Key Management Service** inbound rule in the Overview panel of **Windows Defender Firewall with Advanced Security**. Note the following:

- The policy is currently not enabled (the **Enabled** column says **No**.)
- If enabled, the rule would allow communication (the **Action** column says **Allow**.)

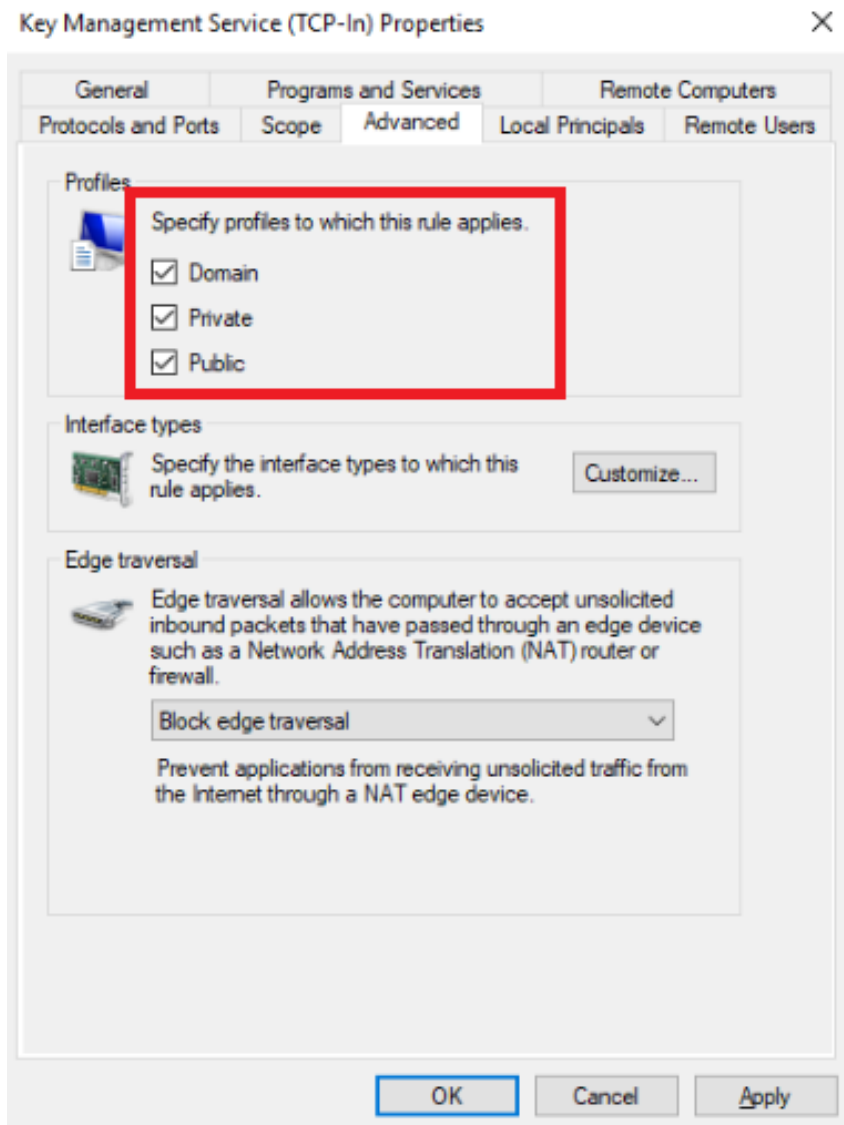
Double-click this rule.



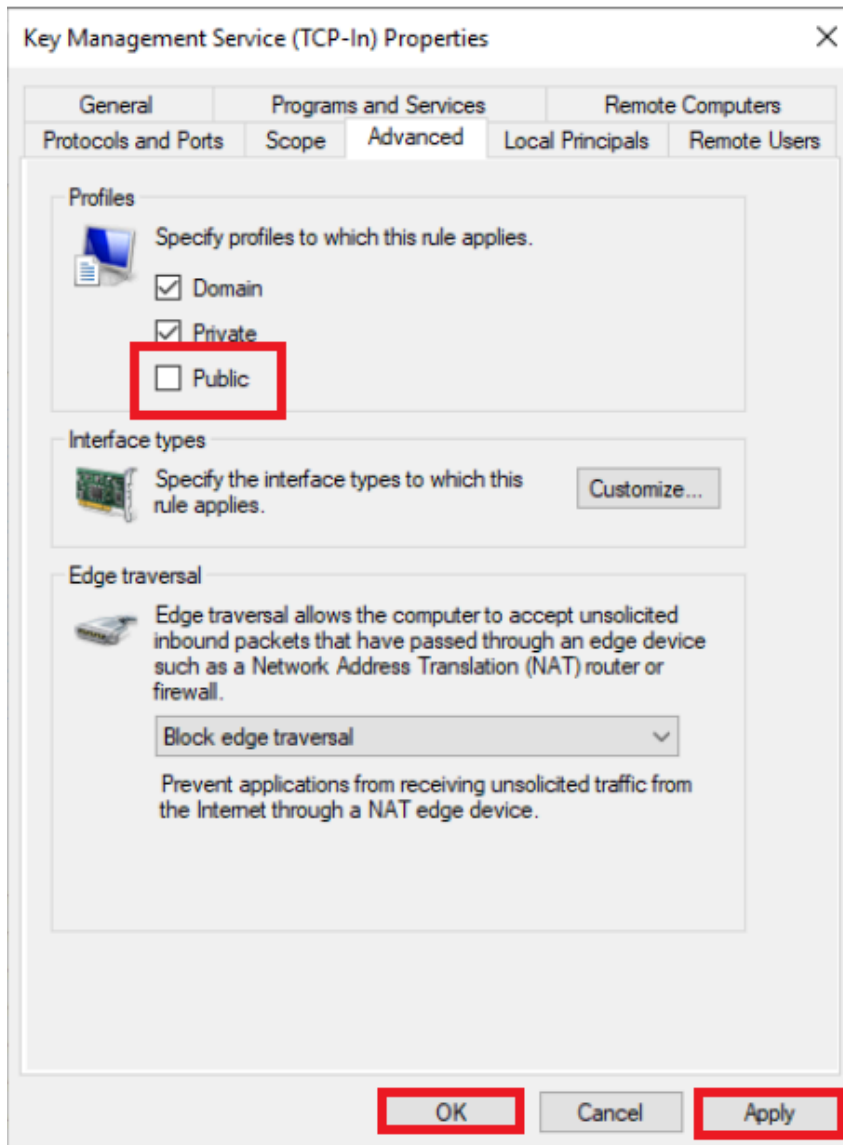
3. Here you will see the details of this rule. You will note that the **General** tab includes the name of the rule, a description of the rule, and whether the rule has been allowed or blocked. In this case, the connection is allowed. Click the **Advanced** tab.



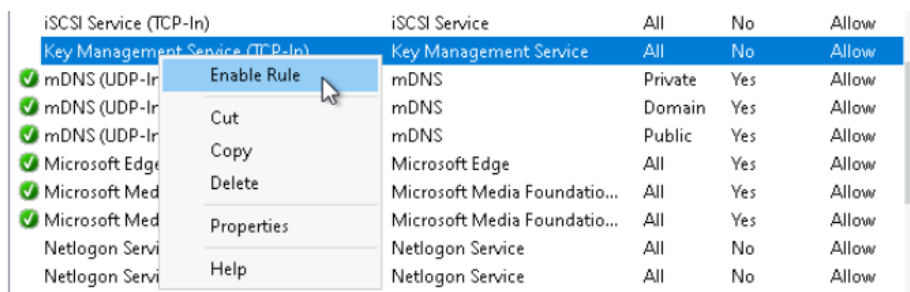
4. Here you will see which profiles the rule applies to. In this case, **Domain**, **Private** and **Public** are all selected.



5. Because we want to allow communication only with the domain and private networks, For **Public** this box should not have a checkmark. Next, click **Apply**, then click **Ok**.



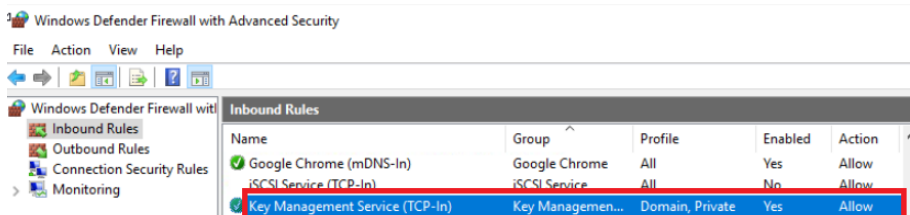
6. Next, right click on Key Management Service(TCP-In) and select Enable Rule.



This will be set as **Yes** which means now the communication with the domain and private network is allowed.



7. Now we will create an inbound rule that blocks communication with the public network. Since the new rule will be similar to the last, we will copy the existing rule. Right-click the **Key Management Service (TCP-In)** inbound rule and click **Copy**. Press **Ctrl+V** to paste.



8. You will now see a second **Key Management Service (TCP-In)** inbound rule. Double-click the second rule to open the **Key Management Service TCP-IN** Properties.

TCP-IP SERVICE (TCP-IN)	TCP-IP SERVICE	API	IPV	ALLOW
✓ Key Management Service (TCP-In)	Key Management Service	All	Yes	Allow
✓ Key Management Service (TCP-In)	Key Management Service	All	Yes	Allow
✓ mDNS (UDP-In)	mDNS	Public	Yes	Allow

9. Since we want to block connection with the public network, select **Block the connection** on the **General** tab. Click **Apply**.

Key Management Service (TCP-In) Properties

Protocols and Ports

Scope

Advanced

Local Principals

Remote Users

General

Programs and Services

Remote Computers

This is a predefined rule and some of its properties cannot be modified.

General

Name:

Key Management Service (TCP-In)

Description:

Inbound rule for the Key Management Service to allow for machine counting and license compliance. [TCP 16881]

Enabled

Action

Allow the connection

Allow the connection if it is secure

Customize...

Block the connection

OK

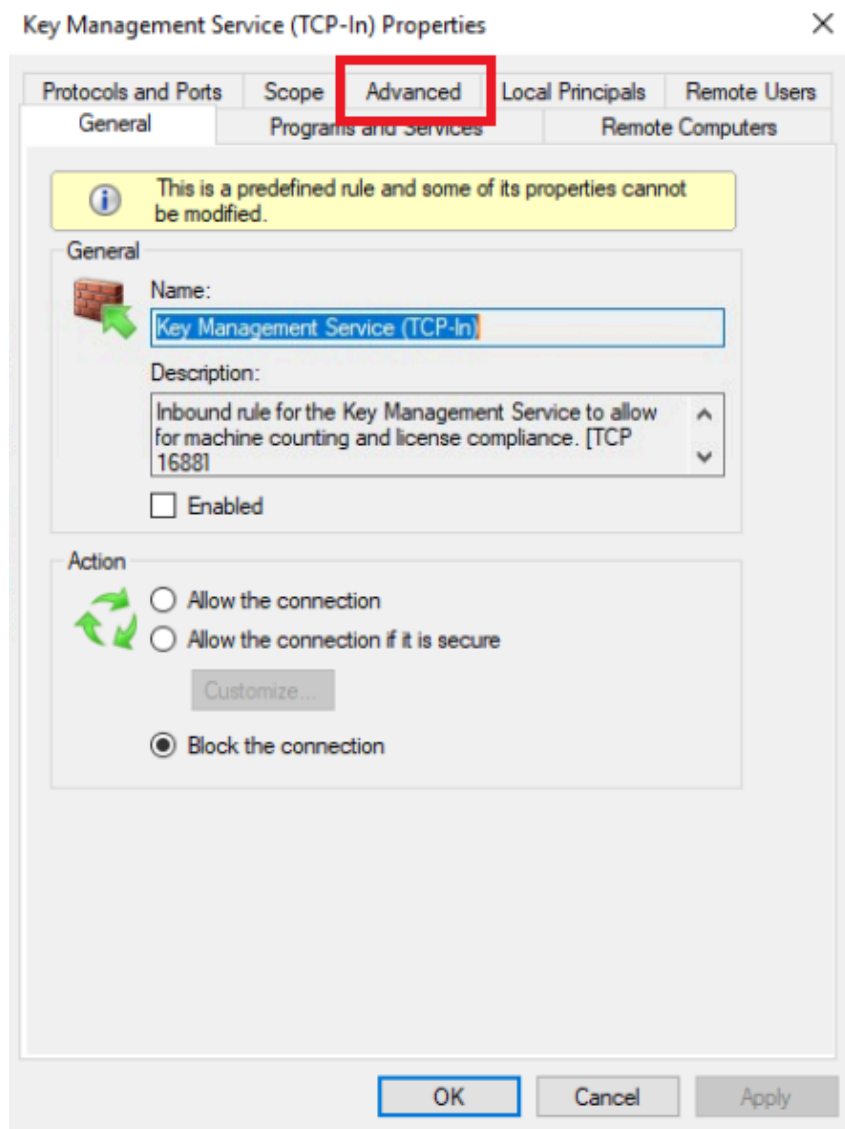
Cancel

Apply

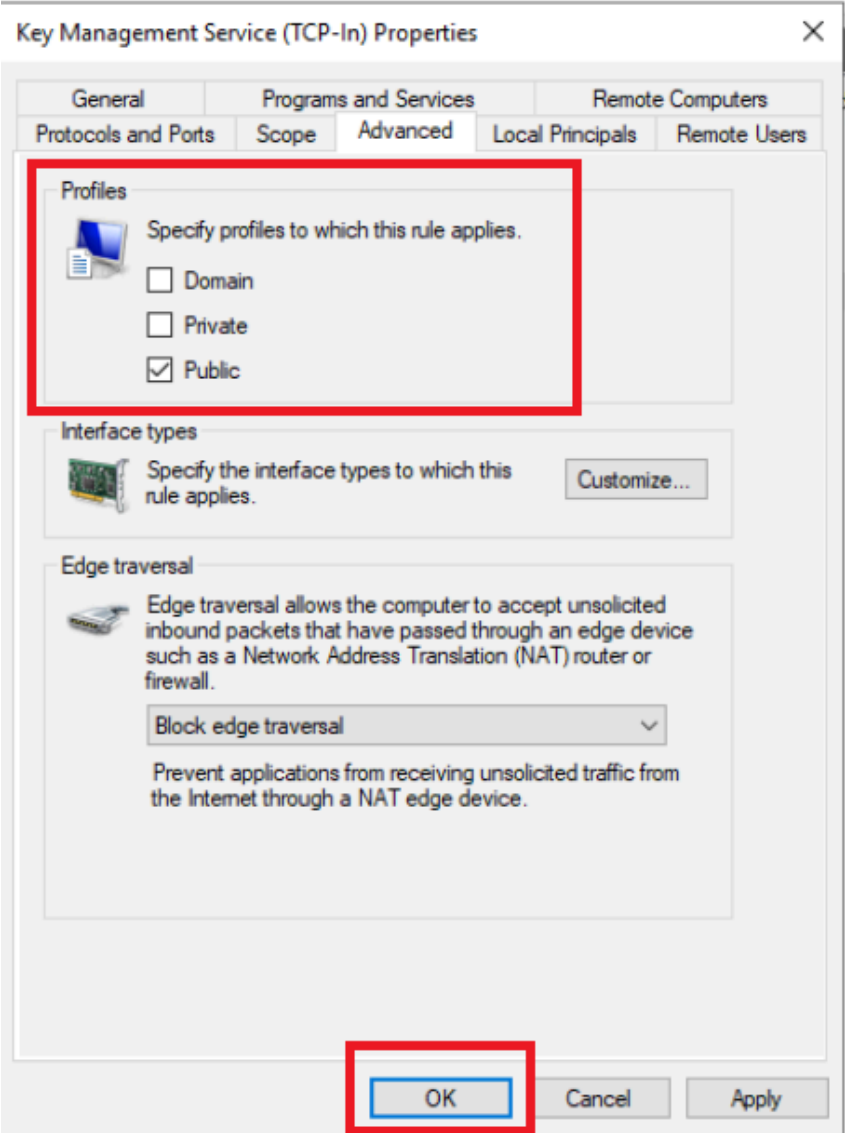
10. Click the **Advanced** tab.

about:blank

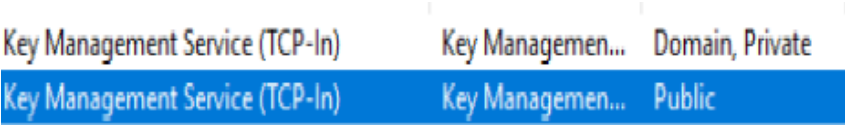
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11. Click the **Domain** and **Private** boxes to remove the checkmarks. Click the **Public** to add the checkmark. Click **Ok**.



12. The Overview panel will show your changes. Right-click each **Key Management Service (TCP-In)** rule and click **Enable rule**.



13. Now you will see that a green checkmark appears next to the first rule indicating that the rule allowing communication is enabled. A circle with a line through it appears next to the second rule indicating that the rule blocking communication is enabled.

Inbound Rules				
Name	Group	Profile	Enabled	Action
✓ Key Management Service (TCP-In)	Key Managemen...	Domain, Private	Yes	Allow
⊗ Key Management Service (TCP-In)	Key Managemen...	Public	Yes	Block

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